

FACULTY MANUAL

Experiment 11

Tableau Dashboard (Basic): Data Import, Preparation and Basic Visualizations

Item	Details
Experiment No.	11
Title	Create dashboards using Tableau: data import, preparation, and basic visualizations
Tool	Tableau Public Desktop (free) or Tableau Desktop (trial / academic licence)
Dataset	Sales_Data.csv (provided, 240 rows)
Duration	2 lab hours (approx. 100-120 minutes)
Level	Basic / Beginner

Note for faculty: Tableau menu names and screen layouts change slightly between releases. Boxes marked *[Insert actual screenshot]* are for you to paste real captures from your lab machine before circulating this manual to students. Expected values (237 orders, approx. 29.6 lakh sales, approx. 5.0 lakh profit) are carried over from the earlier Power BI version of this experiment for the same dataset; please confirm them on your own machine.

1. Aim

To create an interactive dashboard in Tableau by importing a dataset, cleaning and preparing it (data types, data source filters, calculated field), and building basic visualizations (KPI cards, bar chart, line chart, donut chart and filter).

2. Learning Outcomes

- Connect Tableau to a CSV file and review the data on the Data Source page.
- Prepare data in Tableau: check data types, remove blank (null) rows, and create a calculated field.
- Build worksheets (KPI text tables, bar, line and donut charts) by dragging fields to shelves and the Marks card.
- Combine worksheets into a dashboard and add filters to make it interactive.
- Identify and handle common errors (invalid data source, missing fields).

3. What is Tableau?

Tableau is a Business Intelligence (BI) and data visualization tool. It connects to many kinds of data (Excel, CSV, databases, cloud services), and lets users build interactive charts and dashboards by dragging and dropping fields, with no coding needed for basic work.

Tableau is a family of products:

Product	What it is	Used for
Tableau Desktop	Paid desktop application (Windows and macOS)	Full authoring: connecting, preparing, building and saving workbooks privately
Tableau Public (Desktop Public Edition)	Free desktop application (Windows and macOS)	Learning and building dashboards; workbooks can be published to the free Tableau Public cloud or saved locally (used in this experiment)
Tableau Cloud / Server	Online / on-premise sharing platform (licence needed)	Publishing, sharing and scheduling refresh inside an organisation
Tableau Prep Builder	Separate data-preparation tool	Heavier cleaning and combining of data before analysis (not part of this lab)

Product	What it is	Used for
Tableau Mobile	App for Android / iOS	Viewing dashboards published to Tableau Cloud / Server

4. Why Use Tableau?

- **Free to start:** Tableau Public lets students build and save dashboards at no cost.
- **Drag-and-drop analysis:** charts are built by placing fields on shelves, and Show Me suggests suitable chart types.
- **Handles large data:** live connections and extracts stay responsive on far bigger data than a normal spreadsheet.
- **Many data sources:** data from different systems can be combined in one workbook.
- **Interactive:** filters and dashboard actions update every linked sheet at once.
- **Refreshable:** when the source file changes, Data > Refresh updates every view.
- **Cross-platform:** runs on both Windows and macOS.
- **Industry demand:** Tableau skills are widely listed in BI and data analyst job requirements.

5. Applications of Tableau

Domain	Example dashboard
Sales & Retail	Sales by region/product, targets vs actual, top customers
Finance	Revenue, expense and profit tracking, budget variance
Education	Student attendance, result analysis, admissions trends
Healthcare	Patient footfall, bed occupancy, department-wise cost
Manufacturing	Production output, defect rate, machine downtime
HR	Headcount, attrition rate, recruitment pipeline
Marketing	Website traffic, campaign performance, social media reach
Government / Banking	Scheme monitoring, loan and transaction analysis

6. Tableau vs Excel

Feature	Excel	Tableau
Main purpose	Calculations, data entry, small analysis	Visual analytics and dashboards
Data size	Limited (about 10 lakh rows per sheet)	Handles much larger data through extracts and database connections
Building charts	Insert chart, then format cell ranges	Drag fields to shelves; chart type follows from the fields
Interactivity	Basic (filters, slicers, pivots)	Rich: filters, dashboard actions, cross-filtering across sheets
Data refresh	Manual work often needed	Refresh data source; all views update
Sharing	File sharing by email	Tableau Public / Cloud / Server, mobile access
Cost	Paid desktop app (free web version)	Tableau Public is free; Tableau Desktop is paid

Note: Excel is not required for this experiment because the dataset is a CSV file. Excel is useful for opening or editing the CSV, and Tableau can also connect to .xlsx files.

7. Requirements

- PC or laptop with Windows 10/11 or macOS 12 or later, with 4 GB RAM or more (8 GB recommended) and about 1.5 GB free disk space. Unlike Power BI Desktop, Tableau runs on both Windows and macOS.
- Tableau Public Desktop (free) or Tableau Desktop (trial or academic licence).
- Sales_Data.csv, supplied alongside this manual. Copy it to a folder such as C:\Tableau_Lab\ on every machine before the session.

8. Where to Get Excel for Free

Desktop Microsoft Excel is a paid product, but there are legitimate free options. Avoid cracked software or unknown download sites, which often carry malware.

Option	How to get it	Notes
Excel for the web (best free option)	Go to office.com, sign in or create a free Microsoft account, and click Excel.	Runs in the browser; opens/saves .xlsx and .csv. Needs internet; fewer features than desktop.
College / student Microsoft 365 Education	Sign in at office.com with your college email ID (if your institute has a licence).	May include desktop apps depending on the institution. Ask the lab / IT department.
Excel mobile app	Install Microsoft Excel from Play Store / App Store.	Free with limits on small-screen editing.
Microsoft 365 free trial	Only through the official Microsoft 365 trial page on microsoft.com.	About 1 month; a payment method is usually requested. Cancel before it ends.
Free alternatives	LibreOffice Calc or Google Sheets.	Can open and edit CSV files for this experiment.

Faculty tip: Availability and offers change, so verify on microsoft.com before the lab session. Students can simply skip Excel and open the CSV in Notepad if needed.

9. Installing Tableau

Minimum system requirements: Windows 10 or later, or macOS 12 or later; 4 GB RAM or more (8 GB recommended); about 1.5 GB free disk space; internet connection for download and installation.

Method 1: Tableau Public Desktop (free, recommended for students)

1. Open a browser and go to **public.tableau.com/en-us/s/download**.
2. Enter your email address and click **Download the App**.
3. Run the downloaded installer (.exe on Windows, .dmg on macOS). Administrator rights may be needed on lab machines.
4. Accept the licence agreement and click **Install**, then open Tableau Public from the Start menu / Applications folder.
5. You may be asked to sign in to a free Tableau Public account. Signing in is only required if you want to publish to Tableau Public; local saving does not need publishing.

[Insert actual screenshot] Tableau Public download page with the email box and Download the App button.

Important: Tableau Public workbooks are saved either to the free Tableau Public cloud, where they are publicly visible, or, in the current Desktop Public Edition, to the local computer via **File > Save As**. Tell students **not** to publish anything containing private data. Older Tableau Public versions could only save online, so update the lab machines to the latest release.

Method 2: Tableau Desktop (14-day trial or academic licence)

1. Go to the official Tableau website (tableau.com), open the Tableau Desktop trial / download page and download the installer.
2. Run the installer, accept the licence, and click Install.

3. On first launch, start the free trial (about 14 days, full features) or activate with the licence provided by your institute.
4. Institutes with an academic programme licence can issue student keys; ask the lab / IT department.

Faculty tip: Tableau Desktop needs a licence or trial, and trial periods expire, so Tableau Public is the safer choice for a recurring lab. Verify current trial length and academic programme terms on tableau.com before the session.

Installation troubleshooting

Problem	Fix
Installer fails or freezes	Update the OS, run the installer as administrator, temporarily pause third-party antivirus
Installer will not open on macOS	Right-click the .dmg / app and choose Open, or allow the app in System Settings > Privacy & Security
Download blocked by the college network	Ask the administrator to whitelist the Tableau download domains, or use a pre-downloaded installer from a pen drive
Some buttons or dialogs not visible	Set display scaling to 100% and use a screen resolution of at least 1366 x 768
Sign-in prompt appears	Tableau Public accepts a free account; local building works without publishing
"Trial expired" message (Tableau Desktop)	Switch to Tableau Public or ask IT for a licence key

Note: Tableau does not autosave to your computer. Use File > Save As early and File > Save regularly. Tableau also keeps a recovery file, but do not rely on it.

10. Dataset Description

Sales_Data.csv contains sample retail orders. Three rows are deliberately dirty (two blank Region values and one blank Sales value) so students can practise cleaning.

Column	Type	Description
OrderID	Text	Unique order number
OrderDate	Date	Date of order (2025)
Region	Text	North / South / East / West
Category	Text	Electronics / Furniture / Stationery
Quantity	Whole number	Units sold
UnitPrice	Whole number	Price per unit (INR)
Sales	Whole number	Quantity x UnitPrice
Profit	Whole number	Profit on the order (INR)

11. Theory in Brief

- **Workspace:** Tableau has four kinds of pages: the **Data Source** page (connect and preview data), **Worksheets** (single charts), **Dashboards** (several worksheets arranged together) and **Stories** (a sequence of views).
- **Data pane:** fields are split into **Dimensions** (categories such as Region, Category, OrderID) and **Measures** (numbers such as Sales, Profit). Tableau decides this from the data type.
- **Shelves and Marks card:** dragging a field to **Columns** or **Rows** builds the chart axes; the **Marks** card controls Color, Size, Label, Detail, Tooltip and Angle.
- **Pill colours:** blue pills are **discrete** (separate labelled headers); green pills are **continuous** (a continuous axis). A date can be either.
- **Aggregation:** measures are aggregated automatically, for example SUM(Sales). Right-click or use the pill drop-down to change to Average, Count and so on.

- **Live vs Extract:** a live connection reads the file each time; an extract is a saved snapshot inside Tableau. Both leave the original CSV untouched.
- **Filters:** a filter on a sheet limits that sheet's data. A filter can be applied to several sheets. A **data source filter** removes rows before any sheet sees them, which is how this experiment removes the blank rows.

Note: Tableau does not have a recorded "Applied Steps" list like Power Query. Cleaning here is limited to data types, data source filters and calculated fields. For heavier cleaning, Tableau Prep Builder is the companion tool (outside the scope of this basic lab).

12. Procedure (Step by Step)

Part A: Connect to the Dataset

1. Open Tableau Public (or Tableau Desktop). The **Connect** pane appears on the left of the start page.
2. Under **To a File**, click **Text file** (in Tableau Desktop, **More...** may be needed to see it).
3. Browse to **Sales_Data.csv** and click **Open**. The **Data Source** page opens.
4. Review the preview grid at the bottom. Check that the column headers are correct and note the blank (null) cells in Region and Sales.
5. Leave the connection as **Live** (top right) for this experiment. Click **Sheet 1** (bottom left) to go to a worksheet.

[Insert actual screenshot] Start page with Text file highlighted under Connect > To a File.

[Insert actual screenshot] Data Source page showing the 240-row preview grid with null cells in Region and Sales.

Expected result: the Data Source page shows the Sales_Data table with eight fields (OrderID, OrderDate, Region, Category, Quantity, UnitPrice, Sales, Profit).

Part B: Prepare (Clean) the Data

Do steps 1 to 3 on the **Data Source** page, before opening Sheet 1.

1. **Check data types.** Look at the small icon above each column name. OrderDate must show a calendar icon; Quantity, UnitPrice, Sales and Profit must show # (number, whole); OrderID, Region and Category must show **Abc** (string). If wrong, click the icon and choose the correct type.
2. **Remove blank Region rows.** At the top right of the Data Source page, next to **Filters**, click **Add**, then **Add...** in the dialog. Select **Region** and click OK. In the filter dialog, tick **Null** and tick **Exclude**, then click OK.
3. **Remove blank Sales rows.** Repeat: Add > Add... > select **Sales** > OK. Because Sales is numeric, open the **Special** tab and choose **Non-null values**, then OK. Click OK to close the Edit Data Source Filters dialog.
4. **Create a calculated field.** Open Sheet 1. Go to **Analysis > Create Calculated Field...** (or right-click in the Data pane). Name it **ProfitMargin%**, enter the formula **[Profit] / [Sales] * 100**, and click OK. It appears in the Measures area of the Data pane.
5. **Verify the row count.** Double-click **Number of Records** (bottom of the Measures area) so it appears as a number. It should read **237** (240 minus 3 removed rows).
6. **Optional formatting.** Right-click ProfitMargin% > **Default Properties > Number Format...** and choose Number with 1 decimal place.

[Insert actual screenshot] Edit Data Source Filters dialog listing the Region (Null excluded) and Sales (Non-null values) filters.

[Insert actual screenshot] Calculated Field dialog with the ProfitMargin% formula entered.

Expected result: Number of Records = 237, data types are correct and the ProfitMargin% field exists.

Faculty tip: Explain that the original CSV is never modified. Show students that removing a filter from **Edit Data Source Filters** brings the three dirty rows back (Number of Records returns to 240). Also mention that for a ratio across many rows, the better dashboard practice is a field like **SUM([Profit]) / SUM([Sales])** rather than a row-by-row percentage; both are fine for this basic lab.

Part C: Create Basic Visualizations (Worksheets)

Each chart lives on its own worksheet. Add a new one with the **New Worksheet** tab at the bottom. Rename each sheet by double-clicking its tab.

C1. KPI cards (three sheets: Total Sales, Total Profit, Orders)

1. On a new sheet, drag **Sales** onto **Text** in the Marks card. A single number, SUM(Sales), appears. Rename the sheet **Total Sales**.
2. On a second sheet, drag **Profit** onto Text. Rename it **Total Profit**.
3. On a third sheet, drag **Number of Records** onto Text (or drag **OrderID** to Text and change the pill to **Measure > Count (Distinct)**). Rename it **Orders**.
4. On each sheet, click **Text** in the Marks card, then the ... button, to increase the font size and centre the number. Set the **Fit** menu on the toolbar to **Entire View**.

Note: To show sales in lakh, create a calculated field **Sales (Lakh)** = SUM([Sales]) / 100000 and use it on Text. Tableau's built-in number formats do not group digits in the Indian lakh style.

C2. Bar chart: Sales by Region

1. New sheet. Drag **Region** to **Columns** and **Sales** to **Rows** (Show Me will choose a bar chart).
2. Optional: click the **Sort descending** icon on the toolbar. Rename the sheet **Sales by Region**.

C3. Line chart: Monthly trend

1. New sheet. Drag **OrderDate** to **Columns**. Tableau shows YEAR(OrderDate) by default.
2. Click the drop-down arrow on the OrderDate pill and, from the **lower** group of date options, pick **Month** (the continuous, green pill, for example May 2025). Choosing the upper group's Month gives blue discrete month names instead; both are acceptable.
3. Drag **Sales** to **Rows**. Tableau draws a line chart. Rename the sheet **Monthly Trend**.

C4. Donut chart: Sales by Category

Tableau has no built-in donut chart, so a pie chart is built first and a hole is added with a dual axis. If time is short, a plain pie chart (steps 1 to 2) is acceptable.

1. New sheet. In the Marks card drop-down, change **Automatic** to **Pie**. Drag **Category** to **Color** and **Sales** to **Angle**. Set the **Fit** menu to **Entire View**. Optionally drag **Sales** to **Label** and set it to **Quick Table Calculation > Percent of Total**.
2. **Make the hole.** Double-click an empty area of the **Rows** shelf, type **MIN(0)** and press Enter. Do the same once more so there are two MIN(0) pills. The Marks card now has three sections: All, MIN(0) and MIN(0) (2).
3. Build the pie on the **first** MIN(0) section: Pie, with Category on Color and Sales on Angle. (Move the pills over from the All section if needed.)
4. On the **second** MIN(0) (2) section: change the mark type to **Circle**, remove any Color or Angle fields, set **Color** to white, and make the **Size** smaller than the pie.

5. Right-click the second MIN(0) pill on the Rows shelf and choose **Dual Axis**. Right-click the axis and tick **Synchronize Axis** if the circle is off-centre. Right-click each axis and untick **Show Header**.
6. Rename the sheet **Sales by Category**.

C5. Filter (slider): Region

1. Go back to the **Sales by Region** sheet. Drag **Region** to the **Filters** shelf and click OK (all four regions ticked).
2. Right-click the Region pill on the Filters shelf > **Show Filter**. A filter card appears on the right of the view.
3. On the filter card, click its drop-down arrow, then **Single Value (list)** (or leave it as Multiple Values (list)).
4. Click the same drop-down again > **Apply to Worksheets > All Using This Data Source**. Without this, the filter only affects one sheet.

[Insert actual screenshot] Data pane and Marks card showing Region on Columns and Sales on Rows, and the Region filter card.

C6. Build the dashboard

1. Click the **New Dashboard** tab at the bottom (icon with a small grid). Rename it **Sales Dashboard**.
2. In the **Size** box on the left, choose **Fixed size** (for example Desktop Browser 1000 x 800) or Automatic.
3. Drag a **Horizontal** layout container to the top, then drop the three KPI sheets (Total Sales, Total Profit, Orders) into it side by side.
4. Below it, drag **Sales by Region**, **Monthly Trend** and **Sales by Category** into the dashboard. Arrange them in a neat grid.
5. Click the Region filter card on the dashboard, use its drop-down arrow and choose **Apply to Worksheets > All Using This Data Source** if not done already. Move the card to the side.
6. Add a heading: drag a **Text** object to the top, type **Sales Dashboard** and format it. Show sheet titles for each chart by ticking **Show Title** in the sheet's drop-down menu, and edit them to be meaningful.
7. Click the **Sales by Category** chart on the dashboard and click the **Use as Filter** funnel icon at its top right (the dashboard action for Part D).

[Insert actual screenshot] Dashboard tab with the left Objects panel, KPI row at top and charts below.

Expected result: three KPI cards (Total Sales approx. 29.6 lakh, Total Profit approx. 5.0 lakh, Orders = 237), a region-wise bar chart, a monthly trend line, a category donut and a Region filter. Values are for the supplied dataset; verify on your machine.

Part D: Apply Filters and Test Interactivity

1. On the dashboard, click **South** in the Region filter card. All linked sheets and the KPI cards update to show only South-region data (if a chart does not change, its sheet was not set to apply the filter; see Common Student Mistakes).
2. Tick more than one region (Multiple Values list), or click **Only** next to a region name. Select **(All)** to clear the filter.
3. Click a slice of the donut chart. Because **Use as Filter** is on, the other sheets filter to that category. Click an empty part of the chart to clear it.
4. Optional: add a Category filter. On any sheet, drag **Category** to Filters, choose **Show Filter**, apply it to **All Using This Data Source** and select only **Electronics**.

Expected result: cards and all charts recalculate for the selected region or category, confirming interactivity.

[Insert actual screenshot] Dashboard after selecting Region = South.

Part E: Save the Work

1. **File > Save As** and name the file **Exp11_<RollNo>**. Choose **Tableau Packaged Workbook (.twbx)**, which bundles the CSV data with the workbook. In Tableau Public, choose to save to your computer rather than to Tableau Public.
2. (Optional) To submit a printable copy, use **Dashboard > Export Image** (available in both editions) or, in Tableau Desktop, **File > Print to PDF**. Publishing to Tableau Public makes the dashboard visible to everyone, so do not publish unless faculty asks for it.

13. Test Cases

Use these to verify the experiment. Students should record actual results in the last column.

Type	Input	Expected Output	Explanation	Actual
Success	Connect dataset	Data loaded on Data Source page	Validates connection	
Success	Create visuals	Charts displayed	Checks dashboard	
Success	Apply filters	Updated visuals	Tests interactivity	
Failure/Edge	Invalid data source	Connection / file-not-found error	Detects issue	
Failure/Edge	Missing columns	Field flagged with red exclamation mark; visual broken	Checks field mapping	

How to demonstrate the failure / edge cases

Test 4: Invalid data source. Use a workbook saved as a normal **.twb** (or a live connection, not a packaged **.twbx**, which carries its own copy of the data). Close the workbook, rename the CSV (for example **Sales_Dta.csv**) or move it, then reopen the workbook or click **Data > Refresh**. Tableau reports that it cannot find or connect to the file (exact wording varies by version). **Fix:** click **Edit Connection / Locate file** in the message, or **Data > [data source] > Edit Data Source** and reconnect to the correct CSV.

[Insert actual screenshot] Tableau error message when the CSV file cannot be found.

Test 5: Missing columns. In a copy of the CSV, rename the **Profit** header to **Profits**, then click **Data > Refresh** (or reopen the workbook). The Profit field and any sheet that uses it show a **red exclamation mark** and the view fails to draw. **Fix:** right-click the red field and choose **Replace References** (pick the new Profits field), or rename the header back in the CSV.

[Insert actual screenshot] Sheet with a red exclamation mark on the missing Profit field.

Faculty tip: Always restore the original file after demonstrating the failure cases so the next batch is unaffected.

14. Common Student Mistakes and Fixes

Problem	Cause	Fix
Sales appears as text (Abc) and cannot be summed	Wrong data type	Click the data type icon on the Data Source page or in the Data pane and set it to Number (whole)

Problem	Cause	Fix
Chart shows AVG(Sales) or Count of Sales	Wrong aggregation	Click the pill drop-down > Measure > Sum
Dates appear as YEAR only, or as text	Default date level, or date read as a string	Use the pill drop-down and pick Month; if the field shows Abc, change its type to Date
Region filter does not change some charts	Filter applies to one sheet only	Filter drop-down > Apply to Worksheets > All Using This Data Source
Null appears as a bar or filter option	Blank rows not excluded	Add data source filters (Part B) or filter Null out on the sheet
Donut looks like a plain pie	Second MIN(0) pill not set to dual axis / circle	Right-click the second MIN(0) > Dual Axis; set its mark to Circle in white
Clicking the donut does not filter other charts	Use as Filter not enabled	Select the chart on the dashboard and click the funnel icon
KPI card shows a chart, not a number	Field placed on Columns or Rows	Place the field on Text in the Marks card
Work lost after closing	Not saved	File > Save As early and save often

15. Viva Questions

1. What is the difference between a live connection and an extract?
2. What are dimensions and measures? Give two examples from the dataset.
3. What is the difference between a blue (discrete) pill and a green (continuous) pill?
4. What is a calculated field, and how is it different from a column in the CSV?
5. What is the difference between a worksheet and a dashboard?
6. What is the difference between a data source filter and a sheet filter?
7. What does "Use as Filter" do on a dashboard?
8. How do you refresh a workbook when the source file changes?

16. Suggested Evaluation Rubric (15 marks)

Criterion	Marks
Correct data connection and preview on the Data Source page	3
Data preparation: data types, null filters and calculated field	3
At least 4 correct visuals (cards, bar, line, donut) with titles	3
Working filter / dashboard interactivity	3
Failure cases documented and saved file (.twbx)	3

17. Conclusion

Students connected Tableau to a CSV dataset, prepared it with data source filters and a calculated field, built a multi-sheet dashboard with KPI cards, bar, line and donut charts, and verified interactivity using filters and dashboard actions. They also observed how an invalid data source and a missing column affect a workbook.